

Solutions to MP 204 Problem Set 4

(1)

P. 1

This problem gives a rather important result: a magnetic field does no work on a charged particle. That is to say, a particle in a magnetic field will feel a force without gaining any energy. The reason is that we want to show $\frac{dK}{dt} = 0$, which tells us that the particle's kinetic energy is constant. To do this, we use a trick we used in lecture: $\frac{d}{dt} |\vec{v}|^2 = 2\vec{v} \cdot \frac{d\vec{v}}{dt}$.

This is easy to see:

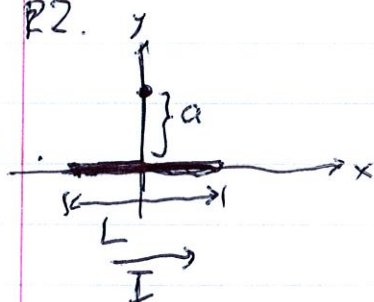
$$\begin{aligned} \frac{d}{dt} |\vec{v}|^2 &= \frac{d}{dt} (v_x^2 + v_y^2 + v_z^2) = 2v_x \frac{dv_x}{dt} + 2v_y \frac{dv_y}{dt} + 2v_z \frac{dv_z}{dt} \\ &= 2(v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \cdot \left(\frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k} \right) \\ &= 2\vec{v} \cdot \frac{d\vec{v}}{dt} \end{aligned}$$

But $\vec{F} = m \frac{d\vec{v}}{dt} = q \vec{v} \times \vec{B}$, so $\frac{d\vec{v}}{dt} = \frac{q}{m} \vec{v} \times \vec{B}$. Thus,

$$\frac{dK}{dt} = \frac{d}{dt} \left(\frac{1}{2} m |\vec{v}|^2 \right) = \frac{1}{2} m \left(2\vec{v} \cdot \frac{d\vec{v}}{dt} \right) = q \vec{v} \cdot (\vec{v} \times \vec{B})$$

But, for any two vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$; thus, $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$. But the dot product of two perpendicular vectors is zero, so $\vec{v} \cdot (\vec{v} \times \vec{B}) = 0$. This proves that $\frac{dK}{dt} = 0$, and thus $K = \text{constant}$. Note that this implies that the particle's speed is constant. Its velocity - as a vector - can change, but its speed, the magnitude of $\vec{v}$, is constant.

P2.



A picture depicting the wire is to the left, with the point $\vec{r} = a\hat{j}$ marked.

(a) The point we're trying to compute $\vec{B}$ at is $\vec{r} = a\hat{j}$, but we also need $\vec{r}'$, the positions that our current elements are located at. Since all points on the wire are on the x-axis, we know $\vec{r}' = x'\hat{i}$, where x' takes values between $-L/2$ and $L/2$. Thus, $\vec{r} - \vec{r}' = a\hat{j} - x'\hat{i}$, so $|\vec{r} - \vec{r}'| = |a\hat{j} - x'\hat{i}| = \sqrt{a^2 + (x')^2}$

and $d\vec{r}' = d(x'\hat{i}) = \hat{i} dx'$. A bit of current between x' and $x' + dx'$ thus contributes a magnetic field of

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{dx' \hat{j} \times (a\hat{j} - x'\hat{i})}{(\sqrt{a^2 + (x')^2})^3} = \frac{\mu_0 I}{4\pi} \frac{a dx' \hat{k}}{[a^2 + (x')^2]^{3/2}} \quad (2)$$

Adding up each of these tiny magnetic fields to get the total field involves integrating this over all values of x' , i.e.

$$\vec{B} = \int d\vec{B} = \int_{-L/2}^{L/2} \frac{\mu_0 I}{4\pi} \frac{a dx' \hat{k}}{[a^2 + (x')^2]^{3/2}} = \frac{\mu_0 I a \hat{k}}{4\pi} \int_{-L/2}^{L/2} \frac{dx'}{[a^2 + (x')^2]^{3/2}}$$

Let's do this integral by making the substitution $x = a \tan \theta$; this is the usual substitution to try whenever the integral has the expression like

$(\text{variable})^2 + (\text{constant})^2$ in it. This substitution is based on the fact

that $1 + \tan^2 \theta = \sec^2 \theta$, so $a^2 + (x')^2 = a^2 + a^2 \tan^2 \theta = a^2 (1 + \tan^2 \theta)$

$= a^2 \sec^2 \theta$. Since $\frac{d}{d\theta} \tan \theta = \sec^2 \theta$, $dx' = a d \tan \theta = a \sec^2 \theta d\theta$.

Finally, at $x' = L/2$, $\tan \theta = \frac{L}{2a} \Rightarrow \theta = \tan^{-1}(\frac{L}{2a})$; similarly,

at $x' = -L/2$, $\theta = -\tan^{-1}(\frac{L}{2a})$. Thus,

$$\begin{aligned} \vec{B} &= \frac{\mu_0 I a \hat{k}}{4\pi} \int_{-\tan^{-1}(L/2a)}^{\tan^{-1}(L/2a)} \frac{a \sec^2 \theta d\theta}{[a^2 \sec^2 \theta]^{3/2}} = \frac{\mu_0 I a \hat{k}}{4\pi} \int_{-\tan^{-1}(L/2a)}^{\tan^{-1}(L/2a)} \frac{d\theta}{a^2 \sec \theta} \\ &= \frac{\mu_0 I \hat{k}}{4\pi a} \int_{-\tan^{-1}(L/2a)}^{\tan^{-1}(L/2a)} \cos \theta d\theta = \frac{\mu_0 I \hat{k}}{4\pi a} \left[\sin \theta \right]_{-\tan^{-1}(L/2a)}^{\tan^{-1}(L/2a)} \end{aligned}$$

To evaluate the limits, recall that $\sec^2 \theta = \frac{1}{\cos^2 \theta} = \frac{1}{1 - \sin^2 \theta} = 1 + \tan^2 \theta$,

or $1 - \sin^2 \theta = \frac{1}{1 + \tan^2 \theta} \Rightarrow \sin^2 \theta = 1 - \frac{1}{1 + \tan^2 \theta} = \frac{\tan^2 \theta}{1 + \tan^2 \theta}$. Thus,

$\sin \theta = \frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$. Since $\tan(\tan^{-1} u) = u$, we have

$$\sin \theta \Big|_{-\tan^{-1}(L/2a)}^{\tan^{-1}(L/2a)} = \frac{\tan(\tan^{-1}(L/2a))}{\sqrt{1 + \tan^2(\tan^{-1}(L/2a))}} - \frac{\tan(-\tan^{-1}(L/2a))}{\sqrt{1 + \tan^2(-\tan^{-1}(L/2a))}}$$

$$= \frac{L/2a}{\sqrt{1 + (L/2a)^2}} + \frac{L/2a}{\sqrt{1 + (L/2a)^2}} = \frac{L}{a \sqrt{1 + \frac{L^2}{4a^2}}}$$

$$= \frac{L}{a \cdot \frac{1}{2a} \sqrt{4a^2 + L^2}} = \frac{2L}{\sqrt{4a^2 + L^2}}$$

Thus, we get our final result:

$$\boxed{\vec{B} = \frac{\mu_0 I L}{2\pi \sqrt{4a^2 + L^2}} \hat{k}}$$

as the magnetic field at the centre of the loop.

⑦

(b) There are many different ways to take the $L \rightarrow \infty$ limit of this, but there's one:

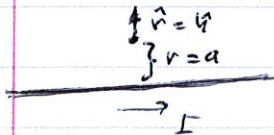
note that $L = \sqrt{L^2} = \sqrt{L^2 + 4a^2 - 4a^2}$, so that

$$\frac{L}{\sqrt{L^2 + 4a^2}} = \frac{\sqrt{L^2 + 4a^2 - 4a^2}}{\sqrt{L^2 + 4a^2}} = \sqrt{1 - \frac{4a^2}{L^2 + 4a^2}}. \text{ As } L \rightarrow \infty, \frac{4a^2}{L^2 + 4a^2} \rightarrow 0,$$

so this fraction approaches 1 as L gets larger and larger. Thus,

$$\lim_{L \rightarrow \infty} \vec{B} = \lim_{L \rightarrow \infty} \frac{\mu_0 I L}{2\pi a \sqrt{L^2 + 4a^2}} \hat{k} = \boxed{\frac{\mu_0 I}{2\pi a} \hat{k}}$$

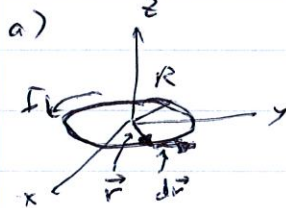
Why is this answer expected? In the $L \rightarrow \infty$ limit, the wire becomes infinitely long, and we already know what $\vec{B}$ is for a current flowing through such a wire: $\frac{\mu_0 I}{2\pi r} \hat{r}$, where r is the perpendicular distance from the wire and $\hat{r}$ is the unit vector pointing directly away from the wire. But $r = a$ is



the distance from the wire, and $\hat{k}$ is the vector pointing away from the wire at this point, so $\frac{\mu_0 I}{2\pi a} \hat{k}$ should be the field, exactly as we found above.

P.3

If you go on to take MP465, you'll meet the magnetic dipole moment again. For now, we just introduce it because it's calculated using the types of integrals we use in the Biot-Savart law and elsewhere, so it's good practice.



The setup is to the left, with $\vec{r}$ pointing out from the wire and $d\vec{r}$ tangent to the loop. But this makes $\vec{r}$ and $d\vec{r}$ perpendicular to one another. Furthermore, $\vec{r} \times d\vec{r}$ points in the z -direction. Thus,

$$\vec{r} \times d\vec{r} = |\vec{r}| |d\vec{r}| \hat{k} = R \hat{k} |d\vec{r}|$$

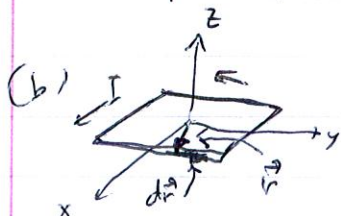
since $|\vec{r}| = R$ everywhere on the loop. This gives

$$\vec{\mu} = \frac{I}{2} \oint_C \vec{r} \times d\vec{r} = \frac{I}{2} \oint_C R \hat{k} |d\vec{r}| = \frac{IR}{2} \hat{k} \oint_C |d\vec{r}|.$$

$|d\vec{r}|$ is just the length of a bit of wire, so $\oint_C |d\vec{r}|$ is the total length, i.e. the circle's circumference, $2\pi R$. Thus,

$$\vec{\mu} = \frac{IR}{2} \hat{k} (2\pi R) = \boxed{\pi R^2 I \hat{k}}$$

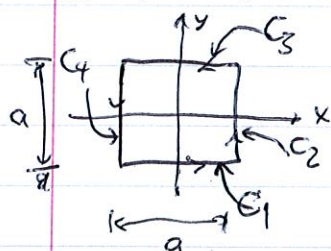
is the loop's magnetic dipole moment.



For this problem, we have the loop to the left. Note that $\vec{r}$ and $d\vec{r}$ are not always perpendicular.

(u)

so we can't use the same trick. However, $\vec{r}$ and $d\vec{r}$ can be found explicitly



by dividing the loop into four parts, C_1, C_2, C_3 and C_4 , as shown.

On C_1 , each part has y-coordinate $-a/2$, and x goes from $-a/2$ to $a/2$. Thus, $\vec{r} = x\hat{i} - \frac{a}{2}\hat{j}$ and $d\vec{r} = dx\hat{i}$.

On C_2 , the x-coordinate is always $a/2$, and they go from $-a/2$ to $a/2$, i.e. $\vec{r} = \frac{a}{2}\hat{i} + y\hat{j}$, $d\vec{r} = dy\hat{j}$.

On C_3 , $y = a/2$ and x starts at $a/2$ and ends at $-a/2$: $\vec{r} = x\hat{i} + \frac{a}{2}\hat{j}$ and $d\vec{r} = dx\hat{i}$.

On C_4 , $x = -a/2$ and y goes from $a/2$ to $-a/2$, i.e. $\vec{r} = -\frac{a}{2}\hat{i} + y\hat{j}$, $d\vec{r} = dy\hat{j}$.

So we can put the integral over all of C into four parts:

$$\begin{aligned}\oint_C \vec{r} \times d\vec{r} &= \int_{C_1} \vec{r} \times d\vec{r} + \int_{C_2} \vec{r} \times d\vec{r} + \int_{C_3} \vec{r} \times d\vec{r} + \int_{C_4} \vec{r} \times d\vec{r} \\ &= \int_{-a/2}^{a/2} (x\hat{i} - \frac{a}{2}\hat{j}) \times dx\hat{i} + \int_{-a/2}^{a/2} (\frac{a}{2}\hat{i} + y\hat{j}) \times dy\hat{j} + \int_{a/2}^{-a/2} (x\hat{i} + \frac{a}{2}\hat{j}) \times dx\hat{i} \\ &\quad + \int_{a/2}^{-a/2} (-\frac{a}{2}\hat{i} + y\hat{j}) \times dy\hat{j} \\ &= \frac{a}{2}k^{\wedge} \int_{-a/2}^{a/2} dx + \frac{a}{2}k^{\wedge} \int_{-a/2}^{a/2} dy - \frac{a}{2}k^{\wedge} \int_{a/2}^{-a/2} dx - \frac{a}{2}k^{\wedge} \int_{a/2}^{-a/2} dy \\ &= \frac{a}{2}k^{\wedge}(a) + \frac{a}{2}k^{\wedge}(a) - \frac{a}{2}k^{\wedge}(-a) - \frac{a}{2}k^{\wedge}(-a) \\ &= 2a^2k^{\wedge}.\end{aligned}$$

Thus, the magnetic dipole moment is

$$\vec{\mu} = \frac{I}{2} \oint_C \vec{r} \times d\vec{r} = \frac{I}{2} (2a^2k^{\wedge}) = \boxed{a^2 I k^{\wedge}}$$

(c) The ratio of $\vec{\mu}/I$ for (a) is $\pi R^2\hat{k}$ and for (b) is $a^2\hat{k}$. Note both point in a direction normal to the plane the loop lies in.

Furthermore, $|\vec{\mu}|/I$ is the area of the region inside the loop - πR^2 for a circle of radius R and a^2 for a square of side length a .

This suggests that, for a planar loop,

$$\boxed{\vec{\mu} = (\text{area of loop}) \times (\text{current in loop}) \times (\text{unit normal to plane loop is in})}$$

which is indeed the correct result.

(5)

P.4

This is just a matter of putting $\vec{E}$ & $\vec{B}$ into the various eq's and seeing what falls out...

(a) Maxwell's 2nd eq'n gives

$$\vec{\nabla} \cdot \vec{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

$$= \frac{\partial}{\partial x} \left(\frac{2E_0}{\omega} y \sin \omega t \right) + \frac{\partial}{\partial y} \left(-\frac{2E_0}{\omega} z \cos \omega t \right) + \frac{\partial}{\partial z} \left(\frac{2E_0}{\omega} x \right)$$

$$= 0 + 0 + 0 \quad \boxed{= 0}$$

so this one works. The third eq'n gives

$$\vec{\nabla} \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} + \frac{\partial B_x}{\partial t} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} + \frac{\partial B_y}{\partial t} \right) \hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} + \frac{\partial B_z}{\partial t} \right) \hat{k}$$

$$= \left[\frac{\partial}{\partial y} (E_0 x z \sin \omega t - y^2 \cos \omega t) \right] - \frac{\partial}{\partial z} (0) + \frac{\partial}{\partial t} \left(\frac{2E_0}{\omega} y \sin \omega t \right) \hat{i}$$

$$+ \left[\frac{\partial}{\partial z} (0) - \frac{\partial}{\partial x} (E_0 x z \sin \omega t - y^2 \cos \omega t) \right] + \frac{\partial}{\partial t} \left(-\frac{E_0}{\omega} z \cos \omega t \right) \hat{j}$$

$$+ \left[\frac{\partial}{\partial x} (0) - \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial t} \left(\frac{E_0}{\omega} x \right) \right] \hat{k}$$

$$= (-2E_0 y \cos \omega t + \frac{2E_0}{\omega} y \cdot \omega \cos \omega t) \hat{i}$$

$$+ (-E_0 z \sin \omega t - \frac{E_0}{\omega} z (-\omega \sin \omega t)) \hat{j} + 0 \hat{k}$$

$$= \vec{0}$$

so $\boxed{\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$ is satisfied.

(b) Maxwell's 4th eq'n can be rewritten as $\rho = \epsilon_0 \vec{\nabla} \cdot \vec{E}$, so we get ρ to be

$$\rho = \epsilon_0 \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right)$$

$$= \epsilon_0 \left(\frac{\partial}{\partial x} (0) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (E_0 (x z \sin \omega t - y^2 \cos \omega t)) \right)$$

$$= \boxed{\epsilon_0 E_0 x \sin \omega t}$$

The fourth eq'n can be rearranged to give

$$\vec{J} = \frac{1}{\mu_0} \vec{\nabla} \times \vec{B} - \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

(6)

so that

$$\begin{aligned}
 \vec{J} &= \left[\frac{1}{\mu_0} \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) - \epsilon_0 \frac{\partial E_x}{\partial t} \right] \hat{i} + \left[\frac{1}{\mu_0} \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) - \epsilon_0 \frac{\partial E_y}{\partial t} \right] \hat{j} + \left[\frac{1}{\mu_0} \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - \epsilon_0 \frac{\partial E_z}{\partial t} \right] \hat{k} \\
 &= \frac{1}{\mu_0} \left[\frac{\partial}{\partial y} \left(\frac{E_0}{\omega} x \right) - \frac{\partial}{\partial z} \left(-\frac{E_0}{\omega} z \cos \omega t \right) \right] \hat{i} + \frac{1}{\mu_0} \left[\frac{\partial}{\partial z} \left(\frac{2E_0}{\omega} y \sin \omega t \right) - \frac{\partial}{\partial x} \left(\frac{E_0}{\omega} x \right) \right] \hat{j} \\
 &\quad + \left\{ \frac{1}{\mu_0} \left[\frac{\partial}{\partial x} \left(-\frac{E_0}{\omega} z \cos \omega t \right) - \frac{\partial}{\partial y} \left(\frac{E_0}{\omega} z y \sin \omega t \right) \right] - \epsilon_0 \frac{\partial}{\partial t} (E_0 x z \sin \omega t - y z \cos \omega t) \right\} \hat{k} \\
 &= \boxed{\left[\frac{E_0}{\mu_0 \omega} \cos \omega t \hat{i} - \frac{E_0}{\mu_0 \omega} \hat{j} + \left(\frac{2E_0}{\mu_0 \omega} + \epsilon_0 E_0 y z \right) \sin \omega t + \epsilon_0 \omega E_0 x z \cos \omega t \right] \hat{k}}
 \end{aligned}$$

(c) Note that J_x and J_y depend only on t ; thus, $\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} = 0$, and

$$\begin{aligned}
 \vec{\nabla} \cdot \vec{J} &= \frac{\partial J_z}{\partial z} = \frac{\partial}{\partial z} \left[-\left(\frac{2E_0}{\mu_0 \omega} + \epsilon_0 E_0 y z \right) \sin \omega t + \epsilon_0 \omega E_0 x z \cos \omega t \right] \\
 &= -\epsilon_0 \omega E_0 x \cos \omega t.
 \end{aligned}$$

Hence,

$$\frac{\partial \phi}{\partial t} = \frac{\partial}{\partial t} (\epsilon_0 E_0 x \sin \omega t) = \epsilon_0 \omega E_0 x \sin \omega t$$

and now see

$$\boxed{\frac{\partial \phi}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0}$$

and so the continuity eq'n is satisfied.